

Riwaq: An Evidence-Bound Computational Framework for Reproducible Hadith Transmission Analysis

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Abstract

Modern academic hadith studies employ several approaches to the history, transmission, and dating of reports. Common-link analysis, introduced by Joseph Schacht and substantially elaborated by G.H.A. Juynboll, examines recurrent branching patterns in *isnād* bundles. Later *isnād-cum-matn* approaches, especially those associated with Harald Motzki, test transmission hypotheses by comparing the structure of transmission chains with variation in the transmitted text. These methods remain debated; no single procedure commands universal acceptance.

Riwaq is a computational research framework designed to make selected structural aspects of transmission analysis explicit, reproducible, and auditable. It represents hadith transmission families as directed graphs, computes transparent structural features, and binds analytical claims to curator-supplied biographical and bibliographic evidence. The numerical score used by Riwaq is a project-specific heuristic operationalization; it is not a statistical probability and is not attributed to Schacht, Juynboll, Motzki, or the classical hadith critics. Likewise, evidence binding establishes traceability between a claim and a source record but does not establish the record's historical truth.

This paper specifies the analytical model, evidence-provenance architecture, release-verification requirements, and a prospective mixed-methods validation study. The paper distinguishes historical and methodological claims from Riwaq-specific engineering choices. Claims about the current software implementation, automated-test counts, release behavior, and deployment state are deferred pending verification against the forthcoming release snapshot.

Keywords: hadith studies; *isnād*; common link; G.H.A. Juynboll; Joseph Schacht; *isnād-cum-matn* analysis; digital humanities; provenance; reproducible research; computational historiography

1. Introduction

1.1 Background

Hadith literature preserves reports attributed to the Prophet Muhammad and, depending on genre and period, to other early Muslim authorities. A report conventionally consists of a *matn*, the reported content, and an *isnād*, the chain of transmitters through which the report is said to have passed. In Sunni legal and hadith traditions, reports attributed to the Prophet became a major vehicle for identifying and transmitting the Prophetic Sunna. Classical Muslim scholars developed extensive methods for evaluating transmitters, transmission relationships, textual variants, and defects in reports.

These methods included biographical evaluation (*al-jarḥ wa-l-ta'dīl*), comparison of transmission routes, and the study of hidden defects (*'ilal*) (Dickinson, 2001; Brown, 2009).

Classical biographical literature preserves information about transmitters' teachers and students, chronological and geographical context, and the judgments of earlier critics. Notable examples include Ibn Abī Ḥātim al-Rāzī's *al-Jarḥ wa-l-Ta'dīl*, al-Mizzī's *Tahdhīb al-Kamāl fī Asmā' al-Rijāl*, al-Dhahabī's *Mīzān al-I'tidāl fī Naqd al-Rijāl*, and Ibn Ḥajar al-'Asqalānī's *Taqrīb al-Tahdhīb*. These works differ in scope and purpose and should not be treated as interchangeable databases.

Modern academic hadith studies developed additional historical-critical approaches. Joseph Schacht introduced the term “common link” into modern hadith scholarship as part of an argument for dating traditions from the structure of their *isnāds* (Schacht, 1950; Aghaei, 2020). G.H.A. Juynboll subsequently elaborated common-link analysis into a more extensive vocabulary of *isnād* bundles, partial common links, single strands, spiders, diving strands, and related patterns (Juynboll, 1983, 2001, 2007). Later scholars have debated the historical interpretations attached to these structural observations.

Harald Motzki and other scholars developed and applied *isnād-cum-matn* analysis, in which hypotheses derived from transmission chains are tested against patterns of agreement and variation in the associated texts. Motzki explicitly described a procedure that collects variants, reconstructs transmission lines, compares textual variants, and then checks whether the results of *isnād* and *matn* analysis correspond before drawing conclusions about transmission history (Motzki, 2000, 2005, 2010).

Riwaq works inside this contested debate without settling it. It does not claim that common-link analysis is the only valid way to study hadith transmission, nor that a structural pattern alone establishes the historical origin of a report. Its purpose is narrower: to make a defined set of analytical assumptions explicit and computationally reproducible so that scholars can inspect, challenge, and compare them.

1.2 Research Problem

Manual transmission analysis presents practical difficulties. An analyst must gather variants, normalize narrator identities, draw or mentally reconstruct transmission relationships, compare branches, consult biographical sources, record uncertainty, and preserve a traceable link between conclusions and evidence. These tasks become harder as variants, transmitters, and source collections multiply.

The problem is not simply speed. A scholarly analysis should make it possible for another researcher to determine:

1. which source witnesses were included;
2. how narrator identities were resolved;
3. how transmission edges were represented;
4. which structural features produced an analytical result;
5. which biographical or bibliographic evidence supports a claim;

6. which assumptions were heuristic rather than historically established; and
7. whether repeated execution on the same data produces the same result.

Riwaq is designed around these requirements.

1.3 Contributions

This paper makes four claims about the framework itself.

First, it provides an explicit computational operationalization of selected common-link and transmission-topology concepts. The operationalization is Riwaq-specific. The numerical weights, normalization rules, penalties, thresholds, and output labels described below are engineering and research-design choices rather than parameters proposed by Schacht, Juynboll, Motzki, or classical Muslim hadith critics.

Second, it specifies a provenance model in which analytical claims are linked to curator-supplied evidence records. The architecture is designed to improve traceability and to prevent unbound claims from being exported as evidence-backed findings. It does not automatically establish the authenticity of the underlying citation.

Third, it defines reproducibility and release-verification requirements for the software. Exact claims about the current implementation, test-suite size, pass status, deployment behavior, and release artifacts must be verified against the release commit that accompanies publication.

Fourth, it proposes an empirical validation program for evaluating whether the framework improves agreement, traceability, usability, or analytical scale in real scholarly workflows.

The paper does not claim priority over every previous computational project involving *isnāds*. Its contribution is this particular combination: explicit structural scoring, evidence provenance, deterministic analysis, and a proposed validation design.

2. Historical and Methodological Background

2.1 Classical Hadith Criticism and Biographical Evidence

Classical hadith criticism cannot be reduced to a single numerical narrator grade. Critics considered multiple dimensions, including a transmitter's integrity (*ʿadāla*), precision or retention (*ḍabt*), the continuity and plausibility of transmission relationships, comparison with other transmitters, and defects that might not be visible from the apparent chain alone. The terminology used by critics also varied by period, critic, and context.

Ibn Abī Ḥātim al-Rāzī (240/854–327/938) preserved extensive material on early hadith critics in the *Taqdima* to *al-Jarḥ wa-l-Taʿdīl* and in the biographical work itself. Dickinson's study of the *Taqdima* shows why critic practice and the intellectual setting of early Sunni hadith criticism matter, and why narrator evaluation is no simple lookup exercise (Dickinson, 2001).

Later biographical compendia systematized large bodies of transmitter information. Al-Mizzī (d. 742/1341) compiled *Tahdhīb al-Kamāl*, a major biographical work on transmitters associated with the

canonical collections. Al-Dhahabī (673/1274–748/1348) produced works including *Mīzān al-I‘tidāl*, which focuses principally on criticized transmitters while also discussing some otherwise respected figures against whom criticism had been recorded. Ibn Ḥajar al-‘Asqalānī (773/1372–852/1449) later condensed and reorganized earlier biographical material in works including *Tahdhīb al-Tahdhīb* and *Taqrīb al-Tahdhīb*. Ibn Ḥajar's *Taqrīb* employs an explicitly ranked evaluative scheme, but those ranks remain the classificatory choices of Ibn Ḥajar within that work; they should not be projected backward as a universal scale used by all earlier critics.

The classical literature also distinguishes methods by which reports or books were received and transmitted. Standard discussions include *samā‘* (hearing directly from a teacher), *qirā‘a* or *‘arḍ* (reading to a teacher), *ijāza* (authorization), *munāwala* (handing over a written text), *mukātaba* (written transmission), *i‘lām* (notification), *waṣiyya* (bequest), and *wijāda* (finding written material) (Ibn al-Ṣalāḥ, 1986). A computational schema that records transmission modality should use historically attested categories, preserve the source's original terminology, and never silently translate every relationship into a single generic edge.

2.2 Common-Link Analysis: Schacht and Juynboll

In modern hadith studies, the term “common link” is first associated with Joseph Schacht. Schacht observed that when multiple *isnāds* of a report are combined, they may converge on a transmitter from whom the chains subsequently branch to multiple later transmitters. He interpreted a significant common link as evidence relevant to the dating and origin of a tradition (Schacht, 1950; Aghaei, 2020).

Juynboll developed this approach much further. Across *Muslim Tradition*, later articles, and the *Encyclopedia of Canonical Ḥadīth*, he constructed and interpreted *isnād* bundles and introduced or elaborated a vocabulary that included partial common links, single strands, spiders, and diving strands (Juynboll, 1983, 2001, 2007). In Juynboll's work, these patterns often carried strong historical implications.

Those implications are contested. The observation that a transmission network branches at a particular narrator is not equivalent to demonstrating that the narrator fabricated, originated, or first publicly disseminated the report. The historical meaning of a common link depends on the completeness of the source corpus, the reliability of the recorded *isnāds*, assumptions about oral and written transmission, and the relationship between textual variants and transmission branches. Riwaq therefore distinguishes the structural detection of branching patterns from historical interpretation.

2.3 Isnād-cum-Matn Analysis

Motzki's *isnād-cum-matn* approach is a methodological counterweight to purely structural dating. In one explicit description of the method, Motzki proposed collecting as many variants as possible, reconstructing transmission lines and potential common links, comparing the texts of the variants, and then testing whether the results of chain analysis and textual analysis correspond (Motzki, 2000). His later survey and methodological work situated this procedure within broader debates about dating Muslim traditions (Motzki, 2005, 2010).

This distinction matters computationally. A graph can identify branching, convergence, and bypass

patterns, but those properties alone cannot establish that reported transmission actually occurred as represented. Correlation between textual variation and transmission branches may strengthen or weaken particular historical reconstructions, but it also requires scholarly judgment about textual similarity, dependence, contamination, paraphrase, and source relationships.

Riwaq therefore treats graph topology as an analytical layer rather than as a self-sufficient historical verdict. A future or optional *matn* module may add text-variant signals, but missing textual analysis must remain missing rather than being silently converted into neutral-looking evidence.

2.4 Reproducibility and Computational Humanities

Computational research is reproducible only when data, transformations, parameters, and outputs can be inspected and rerun. Reproducibility guidance for computational scholarship emphasizes disclosing code, data, workflows, and analysis conditions in enough detail for independent checking (Stodden et al., 2016). Digital history likewise benefits from explicit data models and transparent procedures; historians must stay alert to the interpretive assumptions built into them (Graham, Milligan, & Weingart, 2015).

For Riwaq, reproducibility means more than rerunning code. It requires preserving the evidentiary path from source witness to data representation to analytical output.

3. Riwaq Scope and Design Principles

3.1 Status of the Software Claims in This Manuscript

The present paper specifies the intended analytical and evidentiary behavior of Riwaq. Because the current release code is being updated, this revision does not assert an exact line count, automated-test count, pass count, deployment path, storage key, or repository state. Those are empirical software facts and must be regenerated from the release snapshot used for publication.

The final release should identify, at minimum:

- a repository URL;
- a tagged release or immutable commit hash;
- the software license;
- a machine-generated test report;
- supported runtime environments;
- the data schema version;
- a deterministic example dataset and expected output; and
- instructions sufficient to reproduce the example.

Each item above has a machine-generated source. The release-evidence bundle records them as named fields, the CI test job emits machine-readable test reports, and dedicated commands check the fixture

and deployment artifacts. Appendix A maps each item to its field, artifact, or command.

This separation prevents manuscript prose from drifting away from the software it purports to describe.

3.2 Design Principles

Riwaq follows five design principles.

Local-first analysis. The release target is a client-side workflow in which scholarly data can be processed without mandatory transmission to an external analysis service. Any release that introduces network services or telemetry should state this explicitly and document the data flow.

Deterministic core analysis. Given the same normalized input, configuration, and software version, the core analytical result should be identical. Non-analytical metadata such as export timestamps should be excluded from equality checks.

Explicit assumptions. Feature definitions, weights, thresholds, missing-data rules, and penalties should be visible in the analysis artifact.

Evidence provenance. Assertions about narrators and sources should point to inspectable evidence records containing source and locator information.

Separation of computation from judgment. Riwaq may identify structural patterns and enforce provenance rules, but the software should not present those outputs as final judgments of hadith authenticity.

3.3 Core Data Model

The analytical model distinguishes at least six entity types:

1. **Narrator** — a stable project entity representing a transmitter, with name variants and optional biographical metadata.
2. **Source witness** — a bibliographic or manuscript witness from which a particular transmission or claim is derived.
3. **Variant** — one attested transmission path associated with a specific textual witness.
4. **Transmission edge** — a directed teacher-to-student relation asserted by a variant, optionally annotated with transmission modality and uncertainty.
5. **Transmission family** — a group of variants selected for joint analysis because they are treated as versions of the same report or closely related report.
6. **Evidence record** — a discrete source-backed record supporting a biographical, chronological, or critical claim. Its minimum field set is specified in Appendix B.

A stable narrator identifier is useful for entity resolution, but identifiers are project keys rather than historical evidence. Name normalization must preserve the original source form and the basis on which two references were treated as the same person.

3.4 Graph Representation

For a defined transmission family, Riwaq constructs a directed graph in which nodes represent narrators or collection endpoints and edges represent reported transmission relationships. The conventional direction is from earlier authority or teacher toward later transmitter or collector.

Graph representation is an analytical abstraction. It does not imply that the historical process was a simple directed graph, that every edge represents direct oral contact, or that the surviving corpus is complete. The graph records what the selected sources report, subject to the project's entity-resolution decisions.

Transmission modalities should be represented using source-attested categories where available, such as *samāʿ*, *qirāʾa/arḍ*, *ijāza*, *munāwala*, *mukātaba*, *iʿlām*, *waṣiyya*, and *wijāda*. When the modality is unknown, the system should record it as unknown rather than infer one.

4. Riwaq Structural Analysis

4.1 A Project-Specific Heuristic Model

Riwaq's structural score is a transparent heuristic for prioritizing narrators for scholarly review. It is not a probability that a narrator is a “true” common link, not a measure of hadith authenticity, and not a reconstruction of a numerical method proposed by Juynboll.

The score combines multiple structural signals. Each parameter must be treated as configurable and reported with the analysis. The defaults below are starting values, subject to empirical validation, sensitivity analysis, and expert review.

Let $\text{clamp}(x)$ denote $\max(0, \min(1, x))$.

4.2 Structural Features

4.2.1 Fan-out

$\text{fan_out}(n)$ is the number of distinct direct downstream transmitters of narrator n within the analyzed family:

```
fan_out(n) = |directStudents(n)|
```

High fan-out identifies branching but does not by itself establish a common link in the historical sense.

4.2.2 Variant Coverage

To avoid conflating bundles with individual variants, the feature is called **variant coverage**:

```
variant_coverage(n) =  
    variantsContaining(n) / totalVariants
```

It measures how widely a narrator occurs across the variants in the current analysis.

4.2.3 Collector Diversity

`collector_diversity(n)` counts distinct terminal collection endpoints reachable from `n` under the project's graph representation.

The default normalized signal is:

```
S3 = clamp((collector_diversity(n) - 2) / 6)
```

The values 2 and 8 are engineering normalization anchors, not historically established thresholds. Their effect should be tested in sensitivity analyses.

4.2.4 Pre-Single-Strand Ratio

The **pre-single-strand ratio** is a Riwaq-specific feature describing how much of the selected upstream path structure remains unbranched before the candidate node:

```
pre_single_strand_ratio(n) =  
    singleStrandUpstreamHops(n) / totalUpstreamHops(n)
```

The precise path-selection rule must be specified in the software documentation because graphs may contain multiple upstream routes.

4.2.5 Bypass Ratio

Riwaq uses **bypass** as a neutral computational term for a variant that reaches an earlier node while avoiding a candidate branching node under review.

```
bypass_ratio(n) =  
    bypassVariantsRelativeTo(n) / totalVariants
```

This feature draws on the analytical importance Juynboll attached to alternative and “diving” strands; Riwaq does not assume that every bypass was fabricated or that a bypass has a single historical explanation.

4.2.6 Matn Coherence

When a text-comparison module is active, a `matn_coherence` signal may be computed from explicitly documented text-comparison procedures. When such analysis is unavailable, the value is **missing**. It must not default to 0.50, because “not measured” and “measured as intermediate” are different evidentiary states.

4.3 Default Structural Score

The default signal weights are:

Signal	Default weight	Meaning
S1	0.30	Pre-single-strand ratio
S2	0.25	Variant coverage
S3	0.15	Normalized collector diversity

Signal	Default weight	Meaning
S4	0.20	Normalized fan-out
S5	0.10	Matn coherence, when available

Normalized fan-out is calculated as:

```
S4 = fan_out(n) / max(1, maxFanOutInFamily)
```

The $\max(1, \dots)$ rule prevents division by zero.

If all five signals are available, the structural component is:

```
base =
  0.30*S1 +
  0.25*S2 +
  0.15*S3 +
  0.20*S4 +
  0.10*S5
```

If a signal is unavailable, its weight is removed and the remaining signal weights are renormalized:

```
base =
  sum(w_i * S_i for available signals) /
  sum(w_i for available signals)
```

This rule prevents missing evidence from being converted into an artificial middle score.

4.4 Penalties

Three default penalties are available:

Penalty	Default weight	Meaning
P1	0.20	Bypass ratio
P2	0.10	Detected chronology conflict
P3	0.05	Provenance incompleteness

P1 is the bypass ratio in $[0, 1]$.

P2 is ordinarily 0 or 1 and should be set to 1 only when the available biographical evidence establishes a chronological contradiction under the declared dating assumptions. Missing dates are not themselves chronology conflicts.

P3 is:

```
P3 = 1 - provenance_completeness
```

The final structural support score is:

```
support_score =
```

```
clamp(base - 0.20*P1 - 0.10*P2 - 0.05*P3)
```

The label **support score** is deliberate. It is not a calibrated probability or statistical confidence interval.

4.5 Default Interpretive Bands

For exploratory use, Riwaq may map the support score to configurable descriptive bands:

Label	Default interval	Intended meaning
supported	≥ 0.75	Strong structural support under the declared Riwaq configuration
contested	$0.55 - < 0.75$	Meaningful structural support with unresolved alternatives or contradictions
uncertain	$0.35 - < 0.55$	Insufficient structural support for a strong classification
weak_in_context	< 0.35	Weak support within the analyzed family and configuration

These thresholds are project heuristics. They should not be described as empirical probabilities until calibration data exist.

When two or more candidates remain structurally difficult to distinguish, the software may apply a conservative contradiction rule, but the rule and its threshold must be declared in the analysis artifact. A cap such as 0.70 is an engineering safeguard, not a historically derived constant.

4.6 Worked Example

Assume candidate narrator N has:

- pre-single-strand ratio $S1 = 0.80$;
- variant coverage $S2 = 1.00$;
- collector diversity of 5, giving $S3 = \text{clamp}((5-2)/6) = 0.50$;
- fan-out of 4 where the maximum family fan-out is 4, giving $S4 = 1.00$;
- measured matn coherence $S5 = 0.50$;
- bypass ratio $P1 = 0$;
- no established chronology conflict, $P2 = 0$; and
- complete required provenance, $P3 = 0$.

Then:

```

base =
  0.30(0.80) +
  0.25(1.00) +
  0.15(0.50) +
  0.20(1.00) +
  0.10(0.50)

    = 0.24 + 0.25 + 0.075 + 0.20 + 0.05
    = 0.815

support_score = 0.815

```

Under the default exploratory bands, the candidate would be labeled **supported**. The proper interpretation is: **the candidate receives strong structural support under the declared Riwaq heuristic configuration**. It is not a claim that the narrator historically originated the report. The general reading rule, in both its positive and negative forms, is stated in Appendix C.

4.7 Reliability Information

Classical critical judgments should not be collapsed into a universal numerical reliability scale without an explicit source model. Terms such as *thiqah*, *ṣadūq*, *layyin*, *ḍaʿīf*, and more elaborate phrases have contextual histories, and different critics may disagree.

Riwaq may support an experimental reliability-aware profile, but any such analysis must export:

1. the exact source judgments;
2. the critic and work for each judgment;
3. the edition and locator;
4. the normalization rule, if any;
5. the rule for handling disagreement; and
6. the numerical mapping used in that specific analysis.

No universal 35% reliability weight is asserted in this paper.

5. Evidence Provenance and Claim Integrity

5.1 Evidence Binding Is Not Source Authentication

The evidence architecture addresses a specific computational problem: an exported analytical claim should not appear to be sourced unless it points to a real evidence object in the analysis dataset.

This establishes **referential integrity and provenance traceability**. It does not establish that the quoted source text is correct, that the edition has been cited accurately, that the historical critic's judgment has been interpreted correctly, or that the historical claim is true.

AI-assisted scholarship turns on that distinction. Language models can generate plausible but false citations or source descriptions (Ji et al., 2023). A provenance system can reduce one class of error by

requiring claims to point to pre-existing source records; source verification stays with the scholar.

5.2 Evidence Record

A biographical evidence record should preserve both the source wording and any project normalization. For example:

```
{
  "evidence_id": "ev-hisham-001",
  "narrator_id": "hisham-ibn-urwah",
  "rating": "thiqah",
  "scholar": "Ibn Ḥajar al-ʿAsqalānī",
  "work": "Taqrīb al-Tahdhīb",
  "citation_text": "ثقة فقيه ربما دلس",
  "citation_span": "s.v. Hishām b. ʿUrwa",
  "edition": "Muḥammad ʿAwwāma (ed.), Dār al-Rashīd, 1986",
  "normalization_note": "Normalized to the project's thiqah category; the fuller source wording is retained verbatim in citation_text.",
  "source_type": "print",
  "source_ref": {
    "collection": "Taqrīb al-Tahdhīb",
    "source_type": "print",
    "source_locator": "ʿAwwāma ed. (Dār al-Rashīd, 1986)",
    "ingested_at": "2026-01-15T00:00:00.000Z"
  },
  "ingested_at": "2026-01-15T00:00:00.000Z"
}
```

The wording ثقة فقيه ربما دلس is Ibn Ḥajar's judgment on Hishām b. ʿUrwa in *Taqrīb al-Tahdhīb*.

The example deliberately preserves the source text in `citation_text` separately from the normalized `rating`, with the basis for that normalization recorded in `normalization_note`.

Appendix B specifies the complete minimum field set for an evidence record, including the optional `curated_by`, `revised_at`, and `revision_note` fields this example omits.

5.3 Validation Rules

A release may distinguish among several kinds of problem:

Level	Example condition	Default consequence
blocking	Claim points to a narrator ID that does not exist	Claim excluded from evidence-backed export
blocking	Claim cites an evidence ID that does not exist	Claim excluded from evidence-backed export
blocking	Production artifact contains an explicitly marked fixture/test evidence ID	Claim excluded from production export
warning	Evidence record lacks edition or locator information	Claim retained but marked provenance-incomplete
warning	Source judgments conflict	Disagreement preserved and surfaced

Level	Example condition	Default consequence
info	Additional dissent or contextual note is attached	Flagged for scholarly review

Blocking fixture-like IDs such as `test_*`, `dummy_*`, or `synthetic_*` prevents test-data leakage into production artifacts. It is **not** an authenticity guarantee: a fabricated claim can carry a plausible-looking identifier. Authenticity requires checking the cited source.

5.4 Export Semantics

An evidence-backed export should distinguish at least three statuses:

- **bound** — the claim points to an evidence record with required provenance fields;
- **provenance-incomplete** — an evidence record exists but required bibliographic detail is missing; and
- **unbound** — no qualifying evidence record is attached.

The user should be able to inspect the exact evidence object underlying any evidence-backed claim.

Canonical exports keep the evidence object with the claim: every exported evidence reference includes `evidence_id`, `has_provenance`, and a dereferenced `evidence` object holding the record's bibliographic, normalization, and curation fields (Appendix B). Fields outside that shared projection, including `dissent_notes`, remain in the analysis dataset rather than the export.

6. Release Verification Framework

6.1 Verification and Validation

Software verification asks whether an implementation conforms to its specification. Scholarly validation asks whether the specified and correctly implemented tool is useful and methodologically defensible for its intended research purpose. These are different questions.

A passing software test demonstrates that a tested behavior matched its expected result under the tested conditions. It does not establish the historical validity of the scoring model, the completeness of the source corpus, or the usefulness of the output to hadith scholars.

6.2 Release-Level Verification Requirements

The publication release should include automated checks covering, at minimum:

- schema validation and migration;
- entity resolution and stable identifiers;
- graph construction;
- feature computation;

- missing-data handling;
- chronology rules;
- evidence binding;
- export semantics;
- deterministic output;
- malformed and adversarial inputs; and
- at least one end-to-end analysis from source data to export artifact.

The manuscript should report the test count and pass status **from the release commit**, preferably through a machine-generated CI or test-run artifact. A count belongs in the paper only when it has a reproducible source.

Riwaq's CI `test` job emits a JSON report per supported Node version, uploaded as the `vitest-report-node22` and `vitest-report-node24` artifacts. The release-evidence bundle reads such a report into its `test_counts` field (Appendix A).

6.3 Determinism

Determinism should be tested by executing the same normalized analysis repeatedly and comparing analytical outputs after excluding explicitly non-analytical fields such as timestamps. The release should define which fields are expected to vary and which must remain identical.

Determinism testing should cover at least:

- support scores;
- feature values;
- candidate ordering when ties are resolved;
- evidence bindings; and
- canonical exports.

6.4 Canonical Test Scenarios

High-value fixtures should represent difficult rather than merely trivial cases, including:

- a clear branching candidate;
- a candidate with one or more bypass routes;
- multiple near-equal candidates;
- missing dates;
- established chronology contradictions;
- homonymous narrators requiring entity disambiguation;

- contradictory biographical judgments;
- absent matn analysis;
- empty or malformed variants; and
- evidence records that are structurally valid but provenance-incomplete.

Every end-to-end verification should leave a repeatable artifact such as a canonical JSON report, export file, log, or screenshot. The golden fixture pair provides exactly such an artifact: an input dataset, its expected canonical output, and `npm run verify:golden`, which fails on any mismatch (Appendix A).

6.5 What Verification Cannot Establish

Even complete conformance with the specification would not establish that:

- a common-link interpretation is historically correct;
- the selected corpus is complete;
- a classical judgment has been transcribed correctly;
- a project normalization accurately represents a critic's terminology;
- the chosen feature weights are optimal; or
- scholars will find the tool useful.

Those questions require source criticism, sensitivity analysis, and empirical validation.

7. Proposed Empirical Validation Study

7.1 Objectives

The validation program asks whether Riwaq improves research practice under controlled and real-world conditions. Four questions are proposed.

RQ1: Does use of a shared computational specification change inter-rater agreement on structural common-link/PCL identification compared with manual analysis?

RQ2: Does enforced provenance binding increase the proportion of exported analytical claims with inspectable source support?

RQ3: What training and workflow costs are associated with adoption?

RQ4: Does the framework allow researchers to analyze larger transmission families while preserving auditable evidence trails?

The direction and magnitude of any effect should be treated as empirical questions, not assumed benefits.

7.2 Phase 1: Pilot Controlled Study

A pilot sample of approximately ten scholars may be used to estimate feasibility, variance, task duration, and likely effect sizes. It should not be presented as a definitively powered confirmatory trial unless an a priori power analysis supports that conclusion.

Participants should analyze matched sets of transmission families under two conditions:

1. manual analysis using their normal scholarly workflow; and
2. Riwaq-assisted analysis.

Condition order should be randomized or counterbalanced. The same report should not be analyzed manually and then immediately reanalyzed with Riwaq; carryover and memory effects must be accounted for. Matched but non-identical families, washout periods, or a counterbalanced crossover design can reduce this bias.

Primary measures may include:

- categorical agreement on nominated structural candidates;
- time to a prespecified completion criterion;
- proportion of claims with complete source provenance;
- number and type of unresolved disagreements; and
- System Usability Scale scores or another prespecified usability measure (Brooke, 1996).

If each item is evaluated by exactly two raters, Cohen's κ may be appropriate for categorical agreement. If multiple raters evaluate overlapping sets of items, a multi-rater statistic or hierarchical model should be chosen to match the design. Confidence intervals should accompany agreement estimates.

7.3 Phase 2: Longitudinal Adoption Study

A larger cohort can then use Riwaq in ongoing research for several months. Recruitment targets should be justified by feasibility and the aims of the qualitative component rather than by arbitrary precision.

Data may include:

- opt-in feature-use logs;
- periodic semi-structured interviews;
- researcher-reported workflow problems;
- provenance completeness of exported analyses; and
- instances in which scholars reject or override the structural output.

A local-first architecture creates a privacy advantage but also means usage telemetry cannot be assumed. Study logging should therefore require explicit consent and a documented mechanism, such as participant-initiated export or an ethics-approved study build. No hidden telemetry is necessary for the research question.

7.4 Phase 3: Scale and Stability Assessment

A scale assessment can test whether the software remains usable and computationally stable on hundreds of transmission families. A target such as 500 families is an engineering benchmark rather than a historically meaningful threshold.

Measures may include:

- throughput under a defined curation protocol;
- rates of unresolved entity ambiguity;
- missing-provenance rates;
- frequency of competing high-scoring candidates;
- sensitivity of results to parameter changes;
- computational resource use; and
- proportion of families for which the source corpus is judged sufficiently complete for interpretation.

The aim is not to claim automatic historical discovery at scale but to determine where computational assistance remains reliable and where scholarly curation becomes the limiting factor.

7.5 Ethics

Any study involving identifiable human participants should undergo the ethics review required by the host institution and jurisdiction. Participants should provide informed consent and should be able to withdraw in accordance with the approved protocol.

Study data should be minimized. Research logs, if collected, should be explicitly described, access-controlled, and de-identified or pseudonymized as appropriate. The study should avoid implying that provenance controls eliminate misuse or all forms of fabricated evidence.

8. Discussion

8.1 What Riwaq Formalizes

Riwaq formalizes a limited and inspectable set of operations:

- representation of reported transmission relations as a graph;
- computation of declared structural features;
- application of declared heuristic weights and penalties;
- preservation of evidence provenance; and
- production of reproducible analytical artifacts.

The framework does not formalize the whole of classical hadith criticism, Juynboll's historical

reasoning, or modern ICMA. Those traditions contain interpretive judgments that cannot be reduced without loss to a single graph score.

8.2 Relationship to Schacht, Juynboll, and ICMA

Schacht's common-link argument and Juynboll's elaboration of *isnād* bundles supply historical antecedents for Riwaq's structural layer. Riwaq adopts neither scholar's historical conclusions wholesale.

Motzki's work matters because it shows why transmission topology should be tested against textual variation and broader source coverage rather than interpreted in isolation. Riwaq's current structural layer is therefore one component of a larger research workflow. A mature ICMA-supporting system would need much richer facilities for textual collation, source criticism, uncertainty, and alternative reconstructions.

8.3 Relationship to Classical *ʿIlm al-Rijāl*

Riwaq treats classical biographical works as historical sources to be cited and compared, not as a single homogeneous rating database. The system can store judgments such as *thiqah*, *ṣadūq*, or *ḍaʿīf*, but it should preserve the critic, wording, work, edition, and locator from which the judgment comes.

A computational normalization may help filtering and comparison, but the normalized value must remain secondary to the source wording. Conflicting judgments should stay visible rather than being silently averaged away.

8.4 Evidence Binding and AI-Assisted Scholarship

AI-assisted research creates a particular provenance problem: syntactically convincing citations may be false. Riwaq addresses one part of that problem by requiring evidence-backed claims to reference pre-existing evidence objects.

This architecture should be described accurately. It can prevent an **unbound** claim from masquerading as an evidence-bound claim within the system. It cannot determine on its own whether the source quotation is genuine. Strong source verification still requires consultation of the cited edition or another reliable witness.

8.5 Reproducibility

The principal reproducibility advantage of Riwaq is not that computation is inherently objective. It is that computational rules can be exposed. A scholar can inspect a feature definition, change a threshold, rerun the analysis, and observe whether the conclusion changes.

Sensitivity analysis is therefore essential. If small changes in heuristic weights cause a different candidate to become dominant, the proper scholarly conclusion is not false precision but instability.

8.6 Limitations

The framework has several limitations.

First, surviving *isnād* data are selective historical records, not direct observations of transmission.

Second, entity resolution can be uncertain, especially when names, *kunyas*, *nisbas*, or chronology overlap.

Third, biographical dates may be missing, approximate, or disputed.

Fourth, numerical scores may create an appearance of precision exceeding the evidence. For that reason, Riwaq's scores must remain explicitly heuristic unless empirically calibrated.

Fifth, source coverage strongly affects network topology. Adding early, non-canonical, regional, or otherwise omitted witnesses may substantially change an *isnād* bundle.

Sixth, graph analysis without systematic *matn* comparison cannot reproduce full *isnād-cum-matn* analysis.

Seventh, a provenance record can itself contain a transcription or attribution error. Evidence binding therefore reduces traceability failures but does not eliminate source-criticism failures.

Finally, the empirical utility of the framework remains to be established with scholars using real research tasks.

9. Future Development

Priority areas include:

1. source-aware *matn* collation with explicit variant alignment;
2. sensitivity analysis across weights and thresholds;
3. richer representation of uncertain identities and disputed dates;
4. support for conflicting critic judgments without forced aggregation;
5. multi-scholar evidence curation with version history;
6. temporal and geographical visualization of transmission hypotheses;
7. export of machine-readable provenance graphs; and
8. validation against independently curated scholarly case studies.

Any future API or hosted service should document how it changes the current local-first privacy model.

10. Conclusion

Riwaq is a computational framework for making selected aspects of hadith transmission analysis explicit, reproducible, and evidence-traceable. Its structural layer draws inspiration from common-link and *isnād*-bundle analysis while deliberately separating graph patterns from historical conclusions. Its numerical model is a Riwaq-specific heuristic rather than a method inherited from Schacht, Juynboll, Motzki, or the classical hadith critics.

The evidence architecture binds claims to source records and preserves provenance, but it does not

substitute for verification of the cited source. This distinction is particularly important in AI-assisted scholarly work, where plausible-looking but false citations are a documented failure mode.

The framework should therefore be evaluated at three separate levels: whether the software conforms to its specification; whether its historical source records are accurate; and whether scholars find its outputs methodologically useful. The first requires release-level software verification, the second requires source criticism, and the third requires empirical validation.

The present manuscript specifies those boundaries rather than claiming that they have already been crossed. Once the forthcoming release snapshot is fixed, implementation-specific statements such as test counts, supported environments, deployment behavior, and reproducibility artifacts should be inserted from machine-verifiable release evidence. The release-evidence bundle specified in Appendix A is where that evidence is collected.

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Appendix A. Release Evidence Required Before Submission

Before the manuscript makes implementation-specific claims, archive the following from the release snapshot. Each item is recorded in the release-evidence bundle (node `scripts/release-evidence.mjs --test-report <report.json> > release-evidence.json`):

- immutable commit hash or release tag — bundle fields `commit` and `tag`;
- complete test command — bundle field `test_command`;
- machine-generated test count and pass/fail output — bundle field `test_counts`, read from the vitest JSON report produced by the `test` job in `.github/workflows/static.yml` and uploaded per run as the `vitest-report-node22` and `vitest-report-node24` artifacts;
- list of supported runtime environments — bundle field `engines.node (^22 || ^24)`, exercised by that job's Node 22 and Node 24 matrix;
- schema version — bundle field `schema_version`, read from the import validator (`scripts/json-validator.js`);
- deterministic fixture input and expected canonical output —

tests/fixtures/golden/input.json and tests/fixtures/golden/expected.json, checked by `npm run verify:golden`, with file hashes in bundle hashes.golden_input and hashes.golden_expected;

- deployment artifact — `academic/index.html`, required byte-identical to `index.html` by `npm run verify:parity`, with its hash in bundle hashes.academic_index_html;
- software license — the repository `LICENSE` file, reported in bundle field `license`; and
- repository URL — bundle field `repository`.

The manuscript cites these artifacts rather than restating values that can drift. An archived bundle that reports `dirty: true` or `tag: null` is not release evidence: it must be regenerated from the tagged release commit, and the test report must be attached to the release itself, since workflow-artifact copies expire with Actions retention.

Appendix B. Minimum Evidence Record Fields

A source-backed evidence object should normally include the following. Each point maps to a field of the `ReliabilityEvidence` record in the import schema (authoritative inline definition in `scripts/json-validator.js`, mirrored in `schemas/hadith-record.json`):

- `evidence_id`;
- `narrator_id` or other target entity — `narrator_id`;
- source scholar or author — `scholar`;
- work title — `work`;
- edition or witness — `edition`;
- locator — `citation_span` within the work and `source_ref.source_locator` for the source itself;
- verbatim or faithfully transcribed source text where licensing and workflow permit — `citation_text`;
- project normalization, if used — `rating`, selected from the project's controlled vocabulary, with optional `rating_confidence`;
- normalization note — `normalization_note`;
- uncertainty or dissent notes — `dissent_notes`; and
- curator or revision metadata — `curated_by`, `revised_at`, and `revision_note`.

`edition`, `normalization_note`, `curated_by`, `revised_at`, and `revision_note` are

optional fields added within import schema version 1 without a version bump; the record additionally requires `rating`, `source_type`, `source_ref`, and `ingested_at`. Canonical exports dereference a shared projection of this record (`projectEvidenceRecord` in `scripts/evidence-binding.js`): every exported evidence reference carries `evidence_id`, `has_provenance`, and an `evidence` object holding `scholar`, `work`, `rating`, `citation_text`, `citation_span`, `source_ref`, `edition`, `normalization_note`, `curated_by`, `revised_at`, and `revision_note` — each present as an explicit null when absent — or `evidence: null` when no record is bound. A worked instance appears in §5.2; the field-retention and type rules are exercised in `tests/node/json-validator.test.js`, and the export projection and provenance gate in `tests/node/phase5.test.js`.

Appendix C. Interpretive Rule

A Riwaq result should be read as:

Given the selected source corpus, entity-resolution decisions, declared heuristic configuration, and available provenance records, this candidate exhibits the reported structural pattern.

It should not be read as:

The software has established the historical authenticity, fabrication, or exact date of the hadith.